Football League Database

Data Dictionary

Teams

- team_id: Unique identifier for the team (primary key).
- team_name: Name of the team, for example Data Masters or BI Gods.

Groups

- group_id: Unique identifier for the group (primary key).
- group_name: Name of the organisational group a player belongs to, for example Cohort 4 or HR.

Venues

- venue_id: Unique identifier for the venue (primary key).
- venue_name: Name of the pitch a match is played at, for example Wimbledon 1.

Players

- player_id: Unique identifier for the player (primary key).
- first_name: Player's first name.
- last_name: Player's last name.
- team_id: Identifies the team the player belongs to (foreign key to teams).
- group_id: Identifies the group the player belongs to (foreign key to groups).

Matches

- match_id: Unique identifier for the match (primary key).
- match_datetime: Date and time the match is scheduled or was played.
- home_team_id: Identifies the team playing at home (foreign key to teams).
- away_team_id: Identifies the team playing away (foreign key to teams).
- venue_id: Identifies the venue the match is played at (foreign key to venues).
- match_type: Indicates whether the match is a league fixture or a friendly.
- home_score: Number of goals scored by the home team, left blank until the match has been played.
- away_score: Number of goals scored by the away team, left blank until the match has been played.

Goals

- goal_id: Unique identifier for the goal (primary key).
- match_id: Identifies the match the goal was scored in (foreign key to matches).
- player_id: Identifies the player who scored the goal (foreign key to players).
- minute: Minute of the match the goal was scored in.

Cards

- card_id: Unique identifier for the card (primary key).
- match_id: Identifies the match the card was issued in (foreign key to matches).
- player_id: Identifies the player who received the card (foreign key to players).
- card_type: Indicates whether the card is yellow or red.
- minute: Minute of the match the card was issued in.

Documentation

Goal

The purpose of this task is to develop a relational data model, outlining the entities and the attributes of the data given. We then identify the key relationships (along with associated cardinalities) between entities, visualising the result via an entity relationship diagram. The normalisation process and assumptions made are detailed in this document.

Entities

Entities chosen:

- teams: there are four clubs each with a team made up of consultants and fixtures are played between teams
- groups: the eight cohorts consultants belong to, independent of the team
- venues: the three pitches
- players: the individual consultants who play
- matches: every fixture the tournament produces
- goals: each scoring event occurring within a match
- cards: the yellow/red cards received in a given match

Immediate assumptions:

- fixtures and results can be taken as the same entity (matches). Every row can represent a fixture by date and teams including the 'to be defined' rows
- goals and cards stay separated, as they are distinct events (points in a match versus disciplinary occurrences)

Normalisation analysis

The following section documents the normalisation reasoning applied to each of the seven entities in the schema.

teams

The attribute team_name was repeated as text every time a team was referenced, in both the matches and players tables. This attribute was extracted into its own table to avoid an update anomaly: without this extraction, a change to a team's name would require updating every row in which the name appeared. The primary key team_id follows from the general requirement that every relation possess a unique identifying key.

First Normal Form. No repeating groups or multi-valued attributes are present, and a unique identifying key (team_id) exists. First Normal Form is satisfied.

Second Normal Form. The primary key consists of a single attribute. Second Normal Form is satisfied automatically.

Third Normal Form. Only one non-key attribute (team_name) exists in this relation, leaving no second attribute through which another dependency could occur. Third Normal Form is satisfied.

groups

The attribute group_name was repeated as text every time a group was referenced in the players table and was extracted for the same reason as team_name above. For the same reasons as teams, groups satisfies third normal form.

venues

The attribute venue_name was repeated as text every time a venue was referenced in the matches table and was extracted for the same reason as team_name and group_name above. For the same reasons as teams and groups, venues satisfies third normal form.

players

The original raw data contains no repeating groups, but its given identifier (consultant_id) is duplicated against two different named individuals throughout the dataset and therefore cannot serve as a valid primary key. A surrogate key, player_id, was introduced in its place, and the original name attribute was split into first_name and last_name.

First Normal Form. No repeating groups or multi-valued attributes are present. With the player_id primary key now in place, first normal form is satisfied.

Second Normal Form. The primary key consists of a single attribute, therefore Second Normal Form is satisfied automatically.

Third Normal Form. With four non-key attributes present, a pairwise dependency check was carried out rather than assumed. team_id was checked against group_id using the data directly: Cohort 4 contains players registered to both 'Data Masters' and 'BI Gods', confirming that neither attribute determines the other. first_name and last_name are independent facts about the same individual and are not derivable from one another. No transitive dependency was found; Third Normal Form is satisfied.

matches

The original match data contained two repeating groups - scorers and cards, embedded within each row. These were extracted in full, forming the goals and cards tables described below. A surrogate key - match_id - was introduced rather than relying on the combination of date and teams remaining unique indefinitely.

First Normal Form. Following extraction of the repeating groups, the remaining attributes (date and time, home team, away team, venue, match type, and both scores) are atomic. A unique identifying key exists. First Normal Form is satisfied.

Second Normal Form. The primary key consists of a single attribute. Second Normal Form is satisfied automatically.

Third Normal Form. venue_id was checked against home_team_id: 'Data Masters' as home team is recorded at three different venues across different weeks, confirming no dependency. match_type was checked against match_datetime: although this season's league fixtures occur in 2022 and the friendly fixtures in 2023, this is a coincidence of the current season's data rather than a reliable rule and the attribute was therefore retained as independent. No dependencies between non-key columns were found; Third Normal Form is satisfied.

goals

A surrogate key, goal_id, was used rather than relying on the combination of match, player, and minute remaining unique (as it's possible that the same player scores in a given minute more than once, which would invalidate uniqueness).

First Normal Form. The repeating group has been fully extracted, each row records a single atomic event and a unique identifying key exists.

Second Normal Form. The primary key consists of a single attribute. Second Normal Form is satisfied automatically. This was additionally verified against a hypothetical composite key of match, player, and minute: with three non-key attributes present, no partial dependency would arise even under this alternative structure.

Third Normal Form. match_id, player_id, and minute were checked with each pair in turn: a single match involves many different players scoring at different minutes, and a single player scores across many different matches at different minutes, so no attribute determines another. Therefore, Third Normal Form is satisfied.

cards

First Normal Form. The repeating group has been fully extracted, each row records a single atomic event and a unique identifying key exists.

Second Normal Form. The primary key consists of a single attribute. Second Normal Form is satisfied automatically. It was additionally verified that card_type requires the full combination of match, player, and minute, rather than depending on any single component, confirming no partial dependency would arise even under a hypothetical composite key.

Third Normal Form. card_type was checked against player_id specifically. This could not be conclusively ruled out from the available sample alone, as no player in the dataset received both a yellow and a red card; however, it is ruled out on the basis of domain knowledge, since no football rule permanently associates a specific player with a single card colour. All other attribute pairs were checked and found independent, therefore Third Normal Form is satisfied.

Assumptions

1. Fixture and result are modelled as a single entity, matches, rather than as two separate entities, since every result observed in the data corresponds to exactly one fixture record, confirmed by matching date and teams across both raw sources.
2. Match category - league or friendly - is captured through the match_type attribute rather than through separate entity types.
3. Goals and cards are modelled as two separate entities rather than a single combined event entity, as they measure completely different (distinct) events.
4. A consultant is assumed to belong to exactly one team for the duration of the season. This is modelled as a direct foreign key relationship between players and teams, rather than as a many-to-many relationship.
5. The matches entity relates to teams twice: once for the home role and once for the away role, with both relationships referencing the same entity.
6. League standings, points totals, and goal difference are not modelled as stored data. These are calculated by querying the matches and teams tables directly, rather than being maintained as their own entity. They are not required in the design of the database itself.
7. Goals and cards reference players only. The team associated with a given goal or card is derived through the player's own team assignment, rather than being stored redundantly on the goals or cards tables. This avoids repetition and will save storage space, especially if the data scales to millions of rows.
8. Every table in the database uses a single-attribute surrogate primary key rather than a natural or composite key.
9. The identifying attribute provided in the original raw data - consultant_id - was found to be duplicated against two different individuals and could not serve as a valid primary key. It has been replaced with a newly introduced surrogate key, called player_id.
10. The player's name, given as a single field in the original data, has been separated into first_name and last_name as a composite attribute.
11. Table names are written in the plural and in snake_case, with singular column names throughout as a document preference.

12. The permitted range for the minute attribute is inferred from the values observed in the current season's results, up to a maximum of 90, and may need to be revisited if a different match format or duration is used in a future season.

Constraints (By Entity and Attribute)

The constraints listed below are assumed to hold and can be used upon implementation of this database into a DBMS such as PostgreSQL.

- teams.team_name - unique, not null
- groups.group_name - unique, not null
- venues.venue_name - unique, not null
- players.first_name, players.last_name, players.team_id, players.group_id - not null
- matches.match_datetime, matches.home_team_id, matches.away_team_id, matches.venue_id, matches.match_type - not null
- matches.home_score, matches.away_score - nullable, permitted to be null until the match has been played
- matches.home_team_id and matches.away_team_id - check: home_team_id must not equal away_team_id
- matches.match_type - check: must be one of league or friendly
- goals.match_id, goals.player_id, goals.minute - not null
- cards.match_id, cards.player_id, cards.card_type, cards.minute - not null
- cards.card_type - check: must be one of yellow or red